GELUCIRE® 44/14 FOR TYPE-IV LIPID BASED FORMULATIONS AND THEIR *IN VITRO IN VIVO* PERFORMANCE EVALUATION

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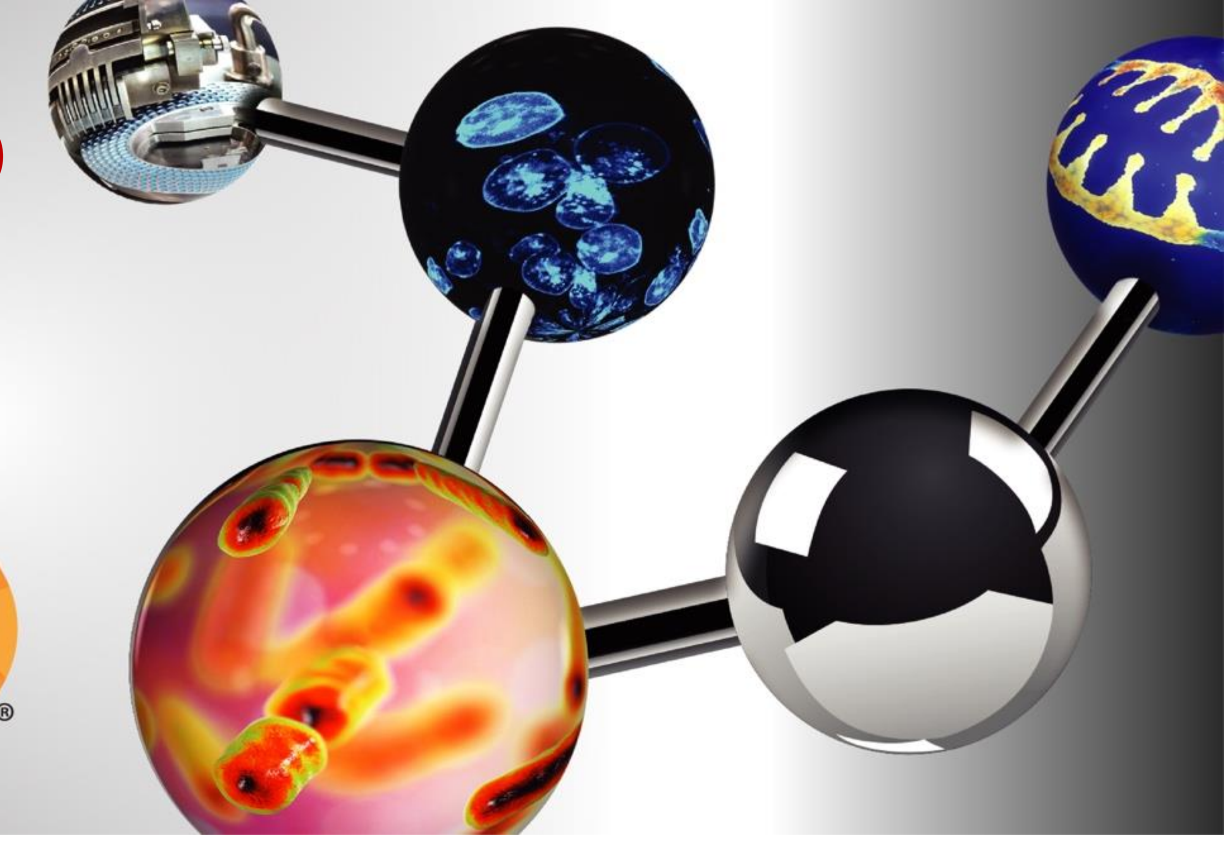

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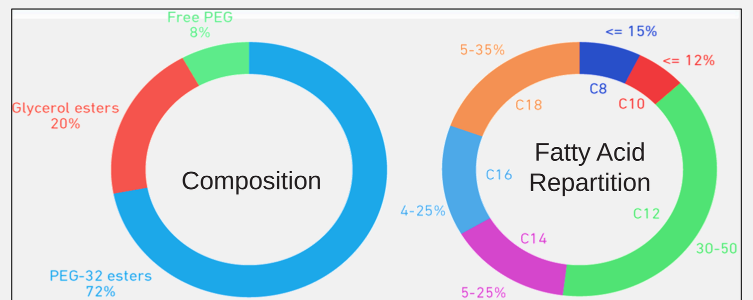
PURPOSE

- Ticagrelor (TGR) is an oral antiplatelet agent, indicated use in thrombotic cardiovascular events.
- TGR belongs to BCS Class-IV drug attributing to its poor solubility and permeability with log P value 2.
- Gelucire® 44/14 (Lauroyl polyoxyl-32 glycerides) is obtained by an alcoholysis reaction between coconut oil and polyethylene glycol-32 (PEG-32) under controlled conditions. It has HLB of 14 and CMC of 70±0.5 mg/L which helps as non-ionic emulsifying base, to improve the pseudo-solubility in GI medium of insoluble drugs.
- Purpose of the present work to evaluate Gelucire® 44/14 as a solubilizer for enhancement of dissolution and *in vivo* performance of TGR.

Properties of Gelucire® 44/14:



Drop point (Mettler, °C)	45.0 ± 2.5
pH (10% in purified water)	4.4 ± 0.7
Relative density	1.023 at 50 °C
Calculated / practical HLB	14 / 11
Critical micellar concentration (g/L, 25 °C)	0.07 ± 0.05
Solubility in ethanol 96 °	Soluble
Solubility in chloroform, methylene chloride	Freely soluble
Solubility in n-hexane	Insoluble

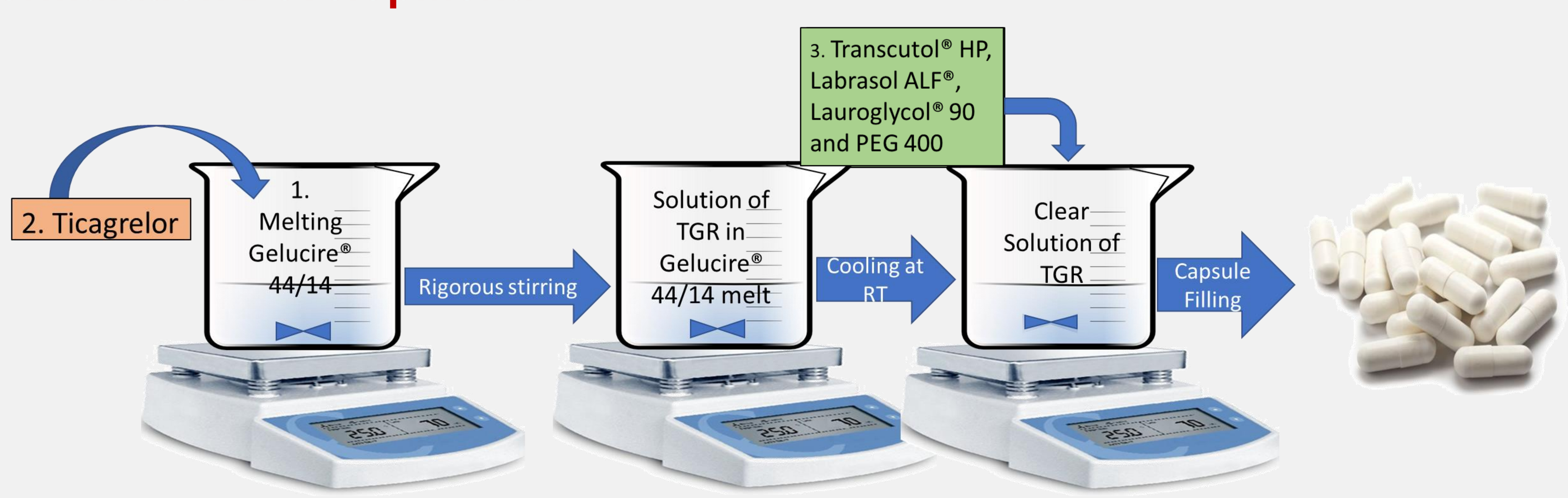
OBJECTIVES

- To prepare capsules of lipid formulations of TGR with Gelucire® 44/14 in combination with auxiliary excipients to meet finally the "Self-Emulsifying Systems" as SEDDS or SMEDDS, for enhancement of the pseudo-solubility and dissolutions of TGR in GI medium.
- To evaluate performance of TGR formulations by *in vitro* lipolysis study and *in vivo* Pharmacokinetic study on rats.

METHODS

Solubility Study: Solubility of TGR was evaluated in water along with it's dispersions with Gelucire® 44/14 in various ratios and other lipid excipients. The samples were stirred for 72h at 25±0.2ºC and evaluated for drug content in supernatant after centrifugation.

Formulation Development:



Step-1: Gelucire® 44/14 was melted at 65ºC (~20ºC above the melting point).

Step-2: TGR added in the molten mass of Gelucire® 44/14 under constant stirring.

Step-3: The solution of TGR in Gelucire® 44/14 cooled till 40ºC followed by addition of Transcutol® HP (Solvent), Labrasol® ALF (cosurfactant), Lauroglycol® 90 (Cosurfactant) and PEG 400 (Cosurfactant). Two formulations were fabricated with different compositions and fill weights i.e. 900 mg (F-1) and 750 mg (F-2).

Step-4: The solution of Step 3 was filled into hard gelatin capsules and kept for cooling at room temperature.

The capsule formulations were evaluated for *in vitro* dissolution tests. *In vitro* lipolysis test and *in vivo* Pharmacokinetic study on rat model.

RESULTS

Solubility Study in Different Excipients:

- The solubility of TGR depicted in the Fig. 1, Gelucire® 44/14 showed enhancement in the solubility in water.
- Water solubility of TGR was found to be 3.58 µg/mL, highest solubility was seen with Transcutol® HP, followed by PEG 400, Labrasol® ALF, Tween® 80, Propylene glycol and Capryol® 90.
- The excipients were selected for formulation based on their TGR solubility and miscibility with Gelucire® 44/14.
- TGR:Gelucire® 44/14 in the ratios of 1:5 and 1:6.6 were taken for further studies.

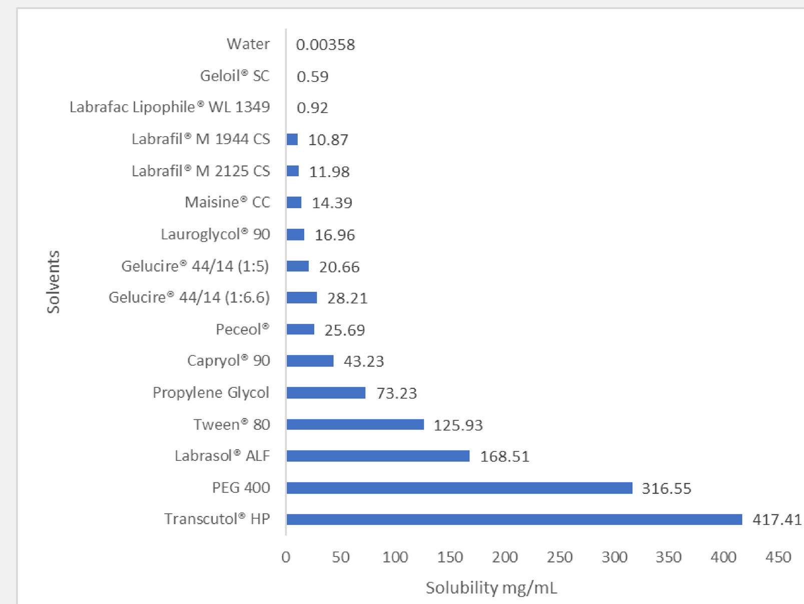


Fig. 1: Solubility of TGR in different excipients.

In vitro Dissolution:

- Dissolution of developed formulations and market reference was performed in HCl Buffer pH 1.2, 0.2% Polysorbate 80 in water, Acetate buffer pH 4.5 and Phosphate buffer pH 6.8, 900 mL, using USP-II apparatus, 75 rpm at 37ºC.
- The dissolution of the developed lipidic formulations was found to be increased remarkably compare to the market reference (Fig. 2).

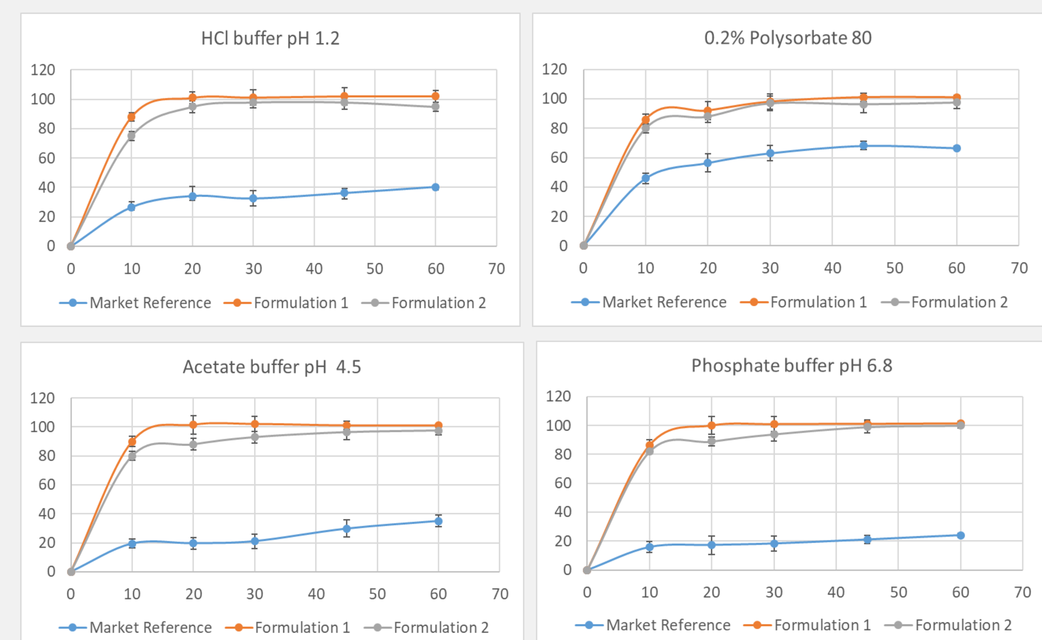


Fig. 2: *In vitro* dissolution of TGR formulations

In vitro Lipolysis Study:

- **Apparatus:** pH stat apparatus (Metrohm AG, Switzerland)
- **Medium:** Lipolysis medium pH 6.5 (fasted state): Tris Maleate 2mM, $CaCl_2$ $2H_2O$ 1.4 mM, NaCl 150 mM, NaTDC 3mM, Phospatidyl choline 0.75 mM, MiliQ water QS.
- **Volume:** 36 mL

Table 1:*In Vitro* Lipolysis:

Parameters	F-1	F-2
Dose (mg)	90	90
Fill weight (mg)	900	750
% dissolution in First 5 min	91.1	89.2
AUC (mg*min)	4316	3915

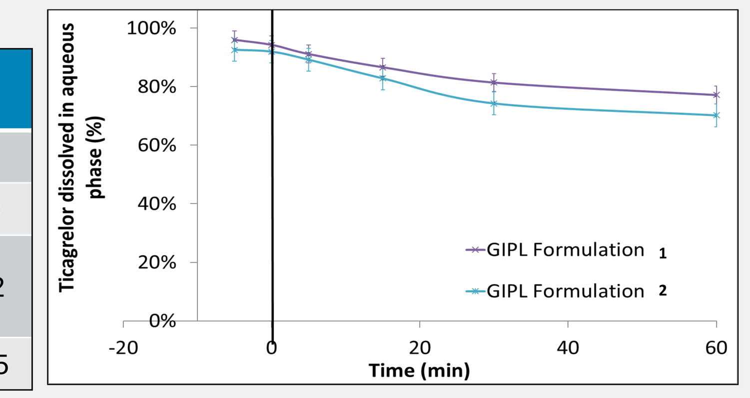


Fig. 3: % Solubility of TGR during *In vitro* lipolysis.

Both Gelucire® 44/14 formulations demonstrated good solubilization capacity to maintain TGR in pseudosolution upon digestion (Table 1, Fig.3).

In vivo Pharmacokinetic Study:

- Animal Model: Wistar male rats with weight of 200 to 250 g
- Dose: 10 mg of TGR per kg body weight by Oral gavage 1 to 1.5 mL
- No. of Groups: 2 groups for GIPL formulations; 1 for Market reference
- Sampling: Blood samples were collected up to 24 h & analyzed by HPLC

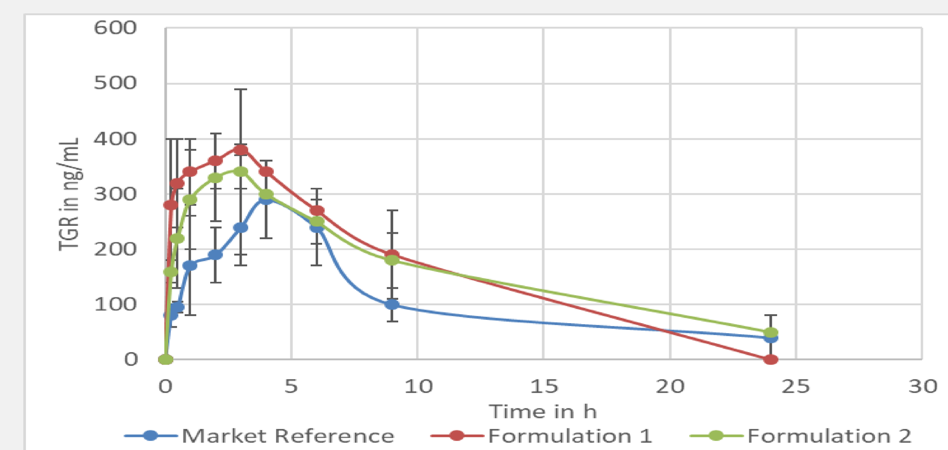


Fig. 4: *In vivo* pharmacokinetic study of TGR formulations.

Table 3: *In vivo* pharmacokinetics of TGR formulations:

Parameters	F-1	F-2	Market reference
T_{max} (h)	3	3	3.54
C_{max} (ng/mL)	380.02	340.51	284.52
AUC_{0-t} (ng/mL*h)	4060.02	4045.85	2845.72

***In vivo* performance of lipidic formulations was found to be significantly better compare to market reference.**

CONCLUSIONS

- Gelucire® 44/14 was employed to effectively enhance the dissolution and *in vivo* performance of a BCS Class-IV drug i.e. TGR.
- *In vitro* lipolysis demonstrates good performance results of developed formulations (obtained during the *in vitro* dissolution test), to maintain stable pseud-solubility of the drug in micelle upon digestion, and finally to predict a good *in vitro in vivo* correlation.
- The *in vivo* pharmacokinetic study on rat model showed significant improvement in the *in vivo* performance of formulations i.e. T_{max}, C_{max}, and AUC; versus the market reference.
- The *in vivo* results obtained are in agreement with the results of *in vitro* lipolysis tests.
- Gelucire® 44/14 can be effectively applied as part of SEDDS or SMEDDS formulations, to enhance dissolution and bioavailability of BCS-Class IV drugs, from Super-Generic (505 b ii) and product lifecycle management perspectives.

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